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The Editor
Financial Innovation

Dear Editor,

I am pleased to submit my manuscript, “**VEIN: On-Chain Causal Risk Attribution for Cryptocurrency Systemic Risk**,” for consideration as a research article in *Financial Innovation*.

The paper proposes VEIN (Verifiable Entity-resolved Interventional/counterfactual Network risk), a framework that treats the resolved on-chain transaction ledger as an auditable candidate structural skeleton for a Pearl-style structural causal model of cryptocurrency systemic risk, rather than inferring the contagion graph statistically from price co-movement as prior causal measures do. On this graph the paper defines an interventional risk measure (On-Chain Causal CoVaR) and, to the best of our knowledge, the first on-chain counterfactual (Pearl Level-3) systemic-risk attribution measure proposed for crypto markets, which decomposes a realised loss into the Shapley-exact share attributable to each upstream entity’s distress. The complete pipeline—a four-tier entity-resolution layer over Dune Analytics’ decoded Ethereum data, the structural causal model, and the risk measures—is implemented and evaluated on real on-chain data around the October 2025 cascading liquidation event, with generality checks on the FTX (2022) and USDC/SVB (2023) stress episodes, and is subjected to a validity and falsification battery (placebo controls, unobserved-confounding robustness at both the coefficient and ranking level, resolution robustness, and external validity against the documented event narrative).

We believe this manuscript is a strong fit for *Financial Innovation*’s scope at the intersection of blockchain, computational financial intelligence, and financial risk management and analysis. The paper combines a methodological contribution (an exact, order-robust counterfactual attribution measure) with an extensive real-data empirical evaluation, and reports negative and suggestive results as candidly as positive ones.

This manuscript is original, has not been published previously, and is not under consideration at any other journal. There are no competing interests to declare, and no external funding supported this work. The complete implementation and all result artefacts underlying the paper are publicly available at <https://github.com/vinhqdang/VEIN-On-Chain-Causal-Risk-Attribution-for-Cryptocurrency-Systemic-Risk> for full reproducibility.

Thank you for considering this manuscript. I look forward to your response.

Sincerely,

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